

## Supporting Information

### Null-Model Choice for Rule-Defined Pattern Counts in Clustered Longitudinal Sequences: A Case Study of Jewish Halachic Menstrual Anticipation Rules

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This document contains the exploratory between-woman pairwise dependence analysis referred to in Sections 3.4, 4.4 and 5.2 of the main article. That analysis is secondary to the null-model comparison and does not bear on the primary question of whether within-woman ordering produces excess rule-defined events. Notation follows the main article; all computations are produced by `scripts/postprocess.py` and the accompanying notebook in the code repository (<https://github.com/dvirross/VesetVectorTest>).

The SHA-256 checksum of the released filtered data file (`data/FilteredData.csv`; 1,554 cycles, 118 women), for verifying file integrity, is

508ee88efb18bcd29c7ed6f841827377bb8d72a8c7dd38fdeb40cf08691f7811

## S1. Exploratory Pairwise Dependence Between Pattern Types

### S1.1 Question and expectations

The joint test of the main article treats the five counts as one vector. The pairwise analysis asks a different and secondary question — whether women who produce one pattern type tend to produce another — as a consistency check against the relationships derived from the pattern definitions (main article, Section 2.2). Those relationships lead to two expectations: a positive  $\text{Haflaga} \times \text{Week-Dilug}$  association, which is partly an arithmetic consequence of containment (every  $\text{Haflaga}$  sequence at  $H = 30$  contains a  $\text{Week-Dilug}$  event) rather than an empirical hypothesis, and a plausible but not entailed positive  $\text{Haflaga} \times \text{Dilug-in-Dilug}$  association from polynomial proximity. No expectation is held for the other eight pairs, so the analysis is a consistency check, not a search.

## S1.2 Methods and rationale

**Unit of analysis: the woman.** — Pattern events are defined within a woman’s sequence and cycles within a woman are not independent, so the woman is the only exchangeable unit for an association test. The analysis works with 118 per-woman summaries, not with 1,554 cycles.

**Rates rather than counts.** — A woman with 45 cycles has roughly nine times as many opportunities to produce *any* pattern as a woman with 5 cycles, so raw per-woman counts of two patterns are positively correlated through follow-up length alone. Dividing each count by the number of cycles removes the first-order part of this exposure effect. It does not remove it entirely: a rate from 5 cycles is far noisier than one from 45, the minimum run lengths (2, 3 or 4 cycles) mean short sequences have proportionally fewer eligible windows, and in these data the Haflaga rate itself correlates with follow-up (Spearman correlation with  $n_i$ : 0.25 for Haflaga, 0.13 for Week, 0.10 for Dilug-in-Dilug, 0.02 for Dilug, 0.00 for Week-Dilug). The permutation null below is therefore one of *marginal* independence between two rates, not of independence conditional on follow-up; Section S1.4 checks how much this matters.

**Spearman rather than Pearson correlation.** — Per-woman rates are bounded at zero, heavily zero-inflated (most women have no event of a given type) and right-skewed. Pearson’s coefficient would be dominated by the few women with several events; the rank correlation measures monotone association and is insensitive to the magnitude of those few values.

**Permutation rather than asymptotic  $p$ -values.** — With 118 women, many tied zeros and skewed rates, the usual  $t$ -approximation for the null distribution of  $r_s$  is unreliable. One rate vector is permuted across women 50,000 times (seed 17); this breaks any association between the two patterns while preserving both marginal distributions exactly, giving a two-sided Monte Carlo  $p$ -value (proportion of permutations with  $|r_s^*| \geq |r_s|$ , with the  $(b+1)/(B+1)$  convention).

**Holm correction over ten pairs.** — Five patterns yield ten pairs, of which only two carry a structural expectation. Holm’s step-down procedure controls the family-wise error rate at  $\alpha = .05$  over all ten and is valid under any dependence among the tests, which matters because the pairs share patterns.

**Binary robustness check.** — Because the rate metric depends on the exposure normalisation and on the few women with several events, the analysis is repeated with each woman coded 1/0 for ever having established each pattern. Each pair is then a  $2 \times 2$  table tested with Fisher’s exact test (conditional on the margins) and with a permutation  $\chi^2$  statistic (permuting one indicator across women), again Holm-corrected. The probability of ever observing an event also increases with follow-up, so this check addresses the choice of metric, not the follow-up dependence.

### S1.3 Results

Table S1 and Figure S1 give the rate-based Spearman correlations. Haflaga  $\times$  Dilug-in-Dilug is positive and survives Holm correction ( $r_s = 0.27$ ,  $p_{\text{perm}} = .004$ , Holm  $p = .038$ ): women with a higher rate of exact repetitions also tend to have a higher rate of second-order progressions, most of which ( $|d| = 1$  in 16 of 27) are near-constant sequences. Haflaga  $\times$  Week-Dilug is positive, as containment requires, but does not survive correction ( $r_s = 0.23$ ,  $p_{\text{perm}} = .011$ , Holm  $p = .10$ ); only 3 of the 26 Haflaga sequences are at  $H = 30$ , so the guaranteed part of the association is small. The remaining eight pairs are null (all Holm  $p = 1$ ), including Dilug  $\times$  Week-Dilug, whose weak negative sign ( $r_s = -0.12$ ) is in the direction implied by their mutual exclusion but is not distinguishable from zero.

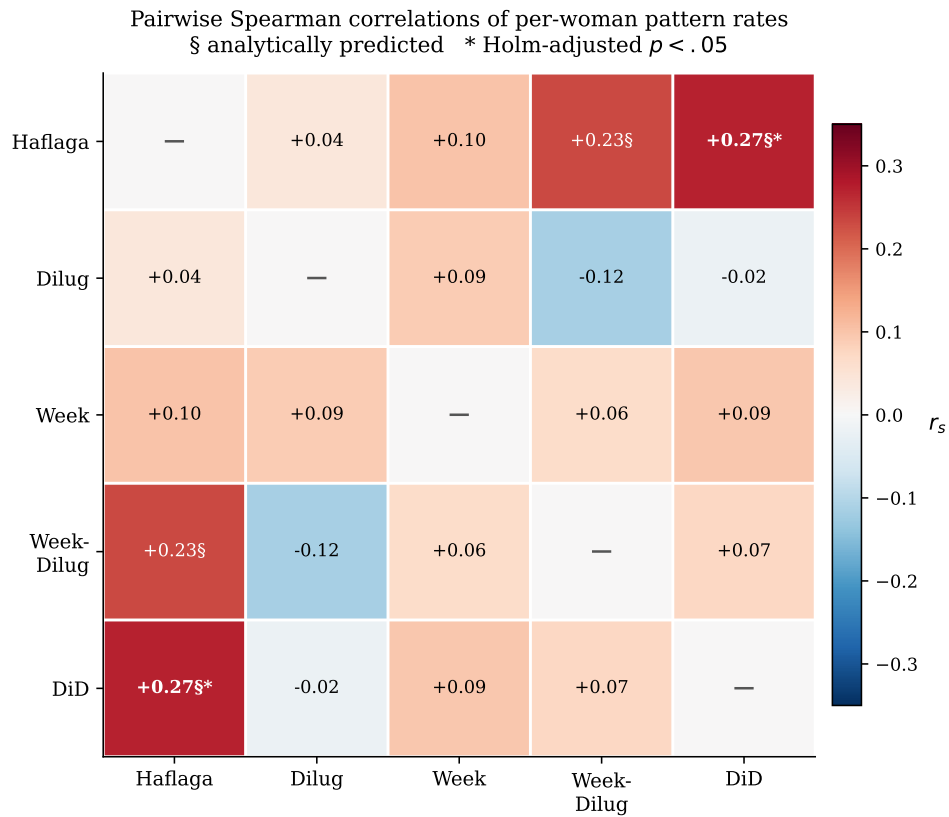
The binary (ever/never) version reproduces the ordering and the two smallest  $p$ -values (Haflaga  $\times$  Dilug-in-Dilug:  $\chi^2 = 9.99$ ,  $p_{\text{perm}} = .004$ , Holm  $p = .042$ , Fisher  $p = .004$ ; Haflaga  $\times$  Week-Dilug:  $\chi^2 = 7.26$ ,  $p_{\text{perm}} = .013$ , Holm  $p = .12$ ), so the findings do not depend on the rate normalisation or on the few women with multiple events.

### S1.4 Robustness to follow-up length

Because both rates in a pair can depend on follow-up, the unrestricted permutation tests marginal rather than follow-up-conditional independence. Two checks, computed by the same script with the same seed, bound the consequence for the two structurally expected pairs. First, the partial Spearman correlation given  $n_i$  (correlation of the rank residuals after regressing each rank vector on the rank of  $n_i$ ) is 0.254 for Haflaga  $\times$  Dilug-in-Dilug (unadjusted 0.269) and 0.239 for Haflaga  $\times$  Week-Dilug (unadjusted 0.231). Second, restricting the 50,000 permutations to

**Table S1:** Pairwise Spearman correlations between per-woman pattern rates ( $N = 118$ ), sorted by  $p_{\text{perm}}$  (two-sided, 50,000 permutations);  $p_{\text{Holm}}$  across the 10 pairs. §: pair with an analytically expected positive association (main article, Section 2.2). Bold:  $p_{\text{Holm}} < .05$ .

| Pair                                               | $r_s$        | $p_{\text{perm}}$ | $p_{\text{Holm}}$ |
|----------------------------------------------------|--------------|-------------------|-------------------|
| <b>Haflaga <math>\times</math> Dilug-in-Dilug§</b> | <b>+0.27</b> | <b>.004</b>       | <b>.038</b>       |
| Haflaga $\times$ Week-Dilug§                       | +0.23        | .011              | .102              |
| Dilug $\times$ Week-Dilug                          | −0.12        | .206              | 1.00              |
| Haflaga $\times$ Week                              | +0.10        | .277              | 1.00              |
| Week $\times$ Dilug-in-Dilug                       | +0.09        | .326              | 1.00              |
| Dilug $\times$ Week                                | +0.09        | .337              | 1.00              |
| Week $\times$ Week-Dilug                           | +0.06        | .476              | 1.00              |
| Week-Dilug $\times$<br>Dilug-in-Dilug              | +0.07        | .480              | 1.00              |
| Haflaga $\times$ Dilug                             | +0.04        | .668              | 1.00              |
| Dilug $\times$ Dilug-in-Dilug                      | −0.02        | .819              | 1.00              |



**Figure S1:** Spearman correlations between per-woman pattern rates. §: analytically expected positive pair; \*: Holm-adjusted  $p < .05$ . All pairwise tests are exploratory.

exchange rates only among women in the same follow-up quintile (strata of 24, 8, 31, 27 and 28 women) preserves each rate's dependence on  $n_i$  while breaking the association between patterns; the resulting two-sided  $p$ -values are .0046 for Haflaga  $\times$  Dilug-in-Dilug (unrestricted .0036) and .0097 for Haflaga  $\times$  Week-Dilug (unrestricted .0113). The exploratory conclusions are therefore not an artefact of unequal follow-up. These checks are reported as robustness diagnostics for an exploratory analysis, not as an additional primary result.

### **S1.5 Interpretation**

Three cautions apply. The associations are between women, not within sequences, and say nothing about temporal ordering. They involve rare patterns (18 Week-Dilug and 27 Dilug-in-Dilug events) and rest on few women. And the surviving association is compatible with the proximity argument but is not evidence for it: attributing co-occurrence to a mechanism would require an explicit generative model for the sequences, which the article does not fit.